CSE 145/237D: Triton Droids

Tactile Hands

Milestone Report

Project Overview

The goal of this project is to develop a low-cost robotic hand prototype with force-sensing resistors (FSRs) on the fingertips of each finger, capable of grasping and releasing simple objects while actively avoiding crushing them. The sensors will be accurate to within 1N. If time allows, additional input methods such as vision-based finger pose estimation, VR teleoperation, and sim-to-real transfer will be explored to extend the system's capabilities.

By the end of the quarter, we will demonstrate a working prototype that can perform basic grasping tasks on a small set of test objects, with the force feedback loop actively modulating grip strength. This project focuses on building a reliable tactile grasping foundation while leaving room for future improvements in teleoperation, simulation, and dexterous manipulation.

Addressing Project Specification Feedback

We received feedback on our Project Specification and made adjustments. Below is a summary of the feedback that was received on our project specification and describes the concrete changes we made in response.

1. Lack of clear team technical expertise

Feedback noted that the project specification did not describe the technical background that each team member brings to their respective subsystem. We addressed this by adding an explicit expertise description with each member's role in the Group Management section. This shows how each subsystem is owned by someone with the background to execute it independently.

2. Poorly defined milestone priorities

Secondly, there was feedback observed that our original milestone list did not distinguish between important items and parallel work streams. It made it difficult to assess schedule risk at a glance. We assigned a single owner to every item. Revising the Milestone Completion table and Gantt chart allows both to reflect this priority structure, making it immediately clear which milestones gate overall progress and which can slip or be cut without affecting the MVP delivery.

MVP Completion Status

Our minimum viable product, as defined in the project specification, is a four-fingered robotic hand with calibrated FSRs accurate to 1N that performs a controlled grasp and release of a fragile object using force feedback. As of Week 6, the MVP is complete. The table below maps each MVP requirement from the project specification to its current status and the evidence supporting completion.

MVP Requirement	Status	Evidence
Operational hand with FSR-based tactile sensing on each fingertip	Complete	Assembled a hand with four FSRs mounted on fingertips, connected via Dupont leads to the prototype board as shown in Figure 3.
Custom prototype board: ESP32-S3	Complete	Board built on a solderable breadboard, mounted on the back of the hand inside a 3D-printed enclosure with status LEDs. Shown in Figure 3.
ESP32 reads force values via voltage dividers and transmits over USB serial	Complete	Firmware streams force values over USB, as shown in Figure 1.

Calibrated force measurement accurate to within 1N	Complete	Power-law fit. Max error 25.7 g over seven loading points, well within the 1N target as shown in Figure 2.
Force aware grasp of at least one simple object without crushing	Complete	The hand grasps a paper cup and stops when measured fingertip force crosses the safe threshold, as shown in figure 3

All MVP requirements are demonstrated and reproducible. The system was shown to be operational during the Week 6 oral presentation.

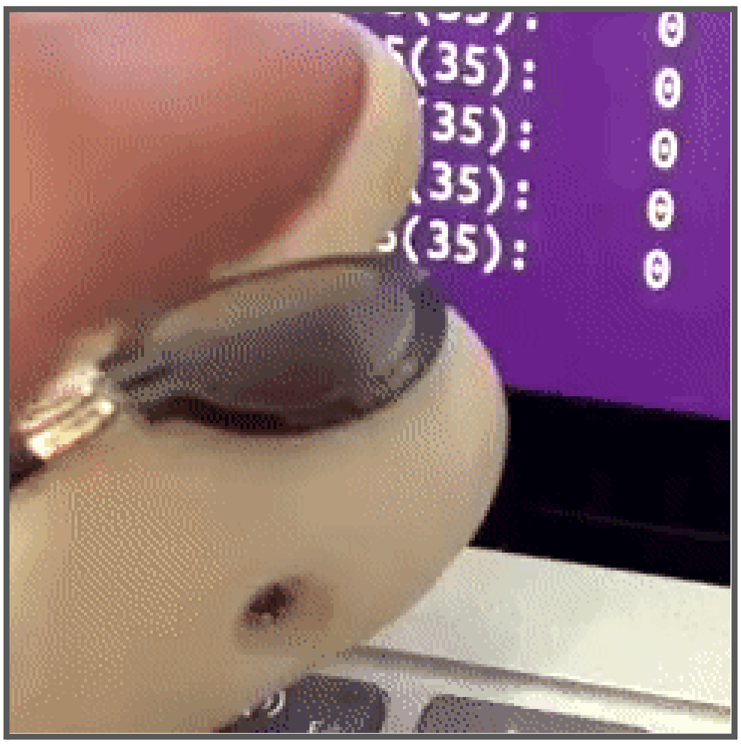

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Figure 1: Live force readings streamed from the ESP32 during the Force Sensing Demo.

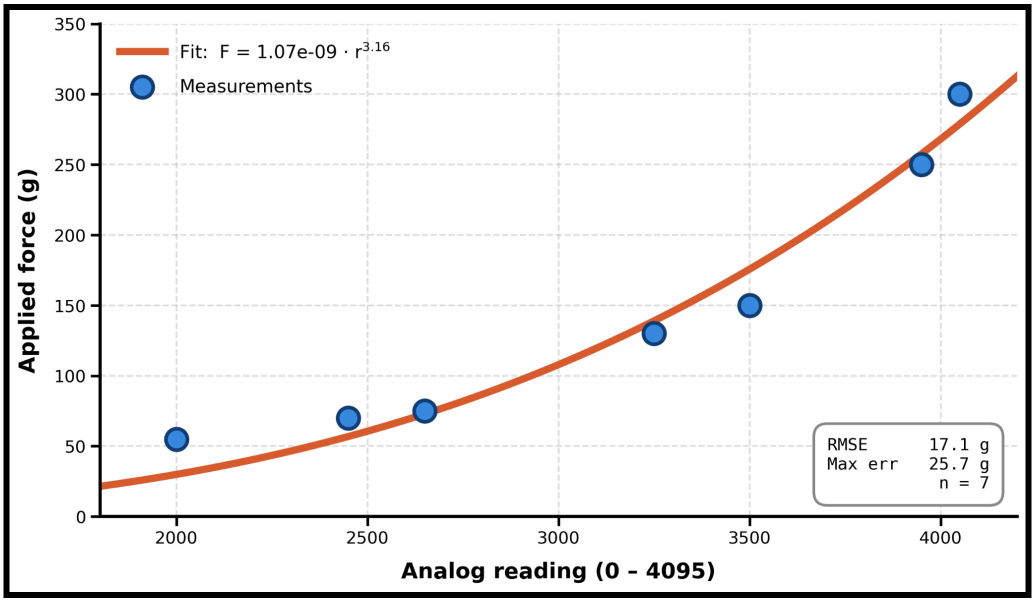

Figure 2: Power law fit for force sensor calibration

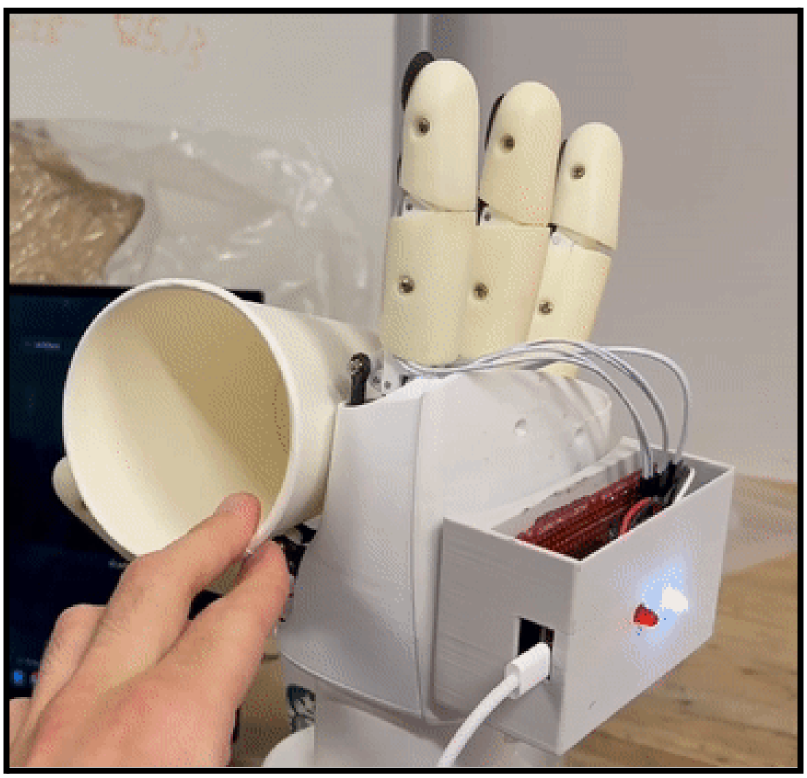

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Figure 3: Force-Aware Grasping Demo. The hand grasps a paper cup and stops when measured fingertip force crosses the safe threshold.

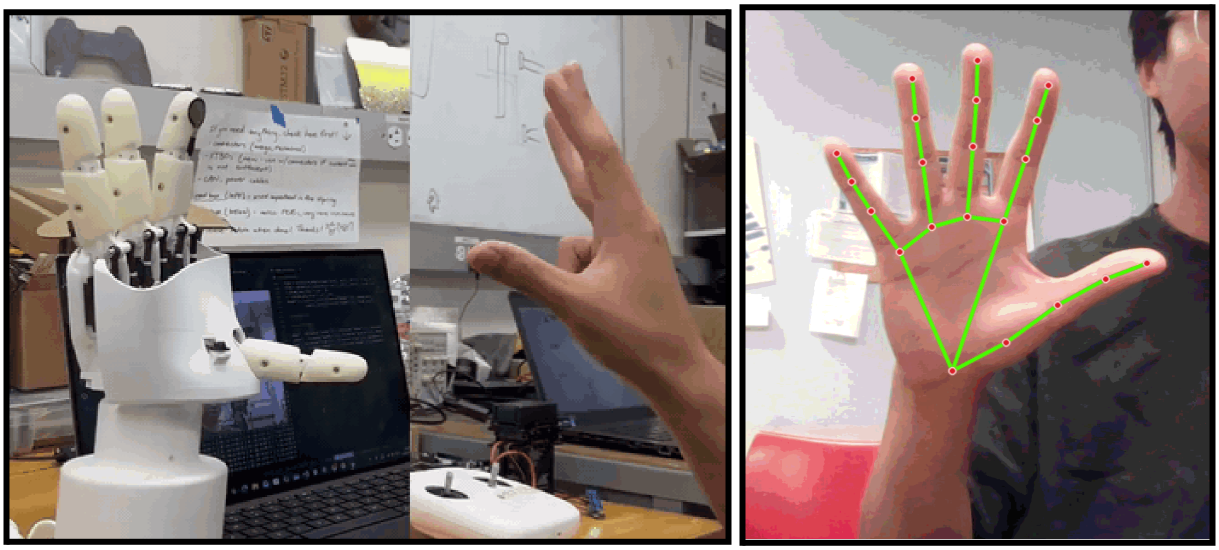
Milestone Completion

The project specification listed milestones culminating in the MVP at Week 6. Below, we document each milestone with its completion date.

Milestone	Target Week	Status	Evidence
System design and component selection	W2-W3	Complete	Hardware bill of materials finalized (ESP32-S3, Pololu FSR, Servos, AmazingHand).
PCB / prototype board design and fabrication	W3-W4	Complete	Solderable breadboard populated with ESP32-S3, voltage dividers, and vertical FSR headers. Shown in Figure 3,
Hand assembly and sensor integration	W3-W5	Complete	All four fingers are actuated by servos; FSRs are mounted at each fingertip. Hand assembly shown in Figure 3.
ESP32 firmware: read FSRs, stream over USB	W4-W5	Complete	Firmware streams are timestamped. The Force Sensing Figure 1 shows live values updating in the terminal as the FSR is pressed.
FSR calibration procedure with 1N accuracy	W4-W6	Complete	Calibration with 3D printed plate to hold various weights; Power-law fit as shown in Figure 2.

Simulation bring-up of the AmazingHand	W3-W5	Complete	Hand model loaded in physics simulator with verified finger kinematics. Hand Simulation Demo Shown in Figure 5.
Testing and force-aware grasping closed loop	W5-W6	Complete	Control loop: detect hand + read FSR ➜ if force limit reached ➜ restrict motion; else send servo command. Demo shown in Figure 3.
MVP completion and validation	W6	Complete	Force-Aware Grasping Demo executed during the Week 6 oral presentation. The hand grasps a paper cup without crushing it.
Hand-pose estimation pipeline (reach goal)	W7-W9	Complete	MediaPipe-based hand-pose detection maps a user's finger curl to servo commands. Figure 3 shows the robotic hand mirroring the user's gesture.
VR controller pose streaming over ROS 2: Valve Index (reach goal, in progress)	W7-W9	Partial	Switched from Quest 2 to Valve Index controllers for easier Python integration (OpenVR/pyopenvr) and native per-finger tracking. Controller position and orientation streaming over ROS 2 is functional. Remaining work: bind pose to ARCTOS arm end-effector control.

Of the ten milestones above, nine are fully complete, and one is partial. Importantly, the partial item (VR controller pose streaming) is itself a reach goal. The partial status reflects that the streaming side is done, but the binding to the ARCTOS arm end-effector remains for Weeks 8-9. Every MVP-track milestone is closed.



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▸ View animated version (right)

Figure 4: Hand Pose Demo. MediaPipe-based finger pose estimation (right) is mapped to the robotic hand mirroring the user’s gesture (left).

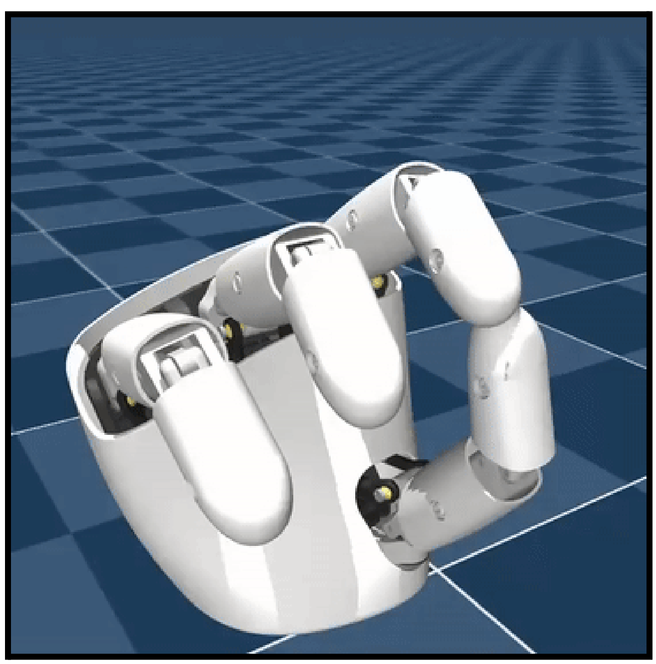

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Figure 5: Hand Simulation. The AmazingHand model loaded into the physics simulator, demonstrating verified finger kinematics and curl-pose rendering.

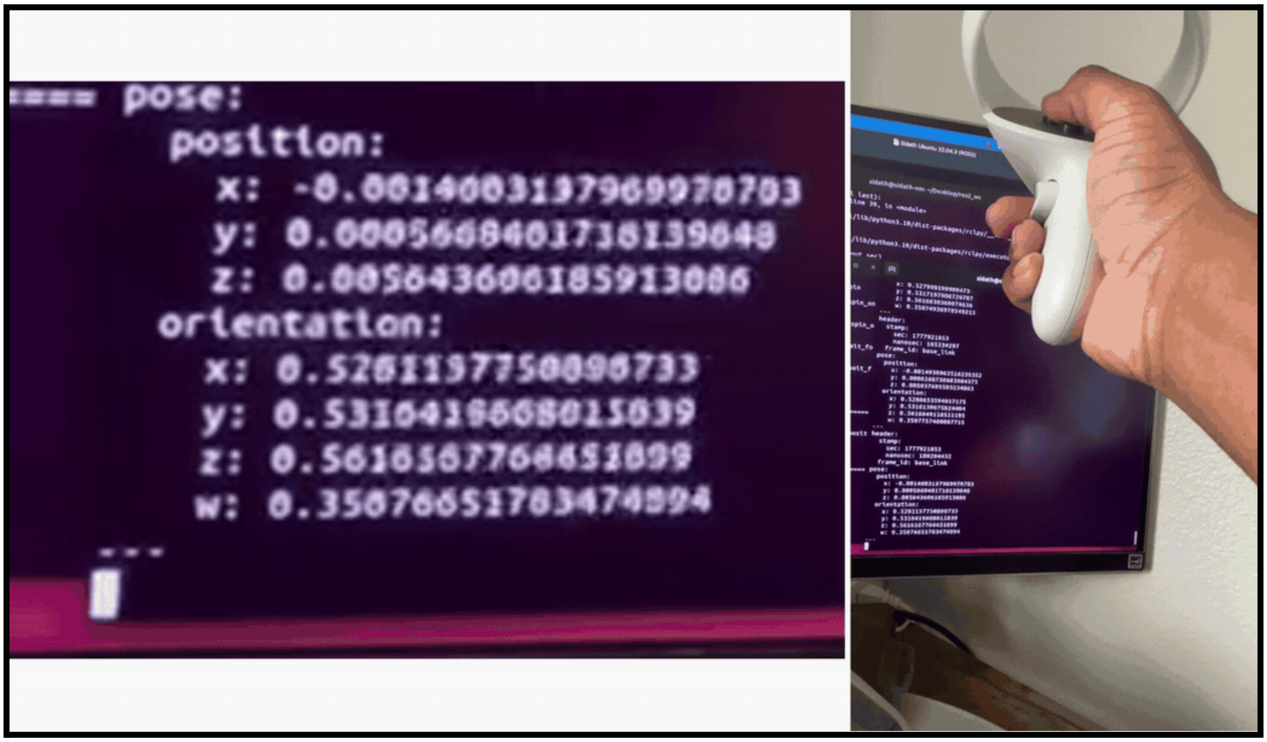

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Figure 6: Virtual Reality Position and Orientation Demo. Quest 2 controller pose (position + orientation) streaming over ROS 2 from the Quest 2 APK on Ubuntu.

Group Management and Per-Member Accomplishments

Each team member owns a subsystem with clearly scoped deliverables. The table below shows what each member has completed to date and what remains for the rest of the quarter. Bryce continues as project lead, coordinating weekly Wednesday standups, tracking blockers via Discord, and maintaining the timeline.

Member	Role	Completed (Weeks 2-6)	Remaining (Weeks 7-10)

Bryce (Project Lead)	Electronics, firmware, project coordination	Designed and built the prototype board; wrote ESP32 firmware, designed and printed the enclosure, fixed ARCTOS wiring. Tracked schedule and ran weekly standups.	Lead final integration of VR teleoperation with the force-aware controller.
Shree	Force sensor calibration	Built the calibration; performed seven-point loading with stacked weights on a PLA plate; fit and validated the power-law model.	Repeat the calibration procedure across all four fingers to characterize finger-to-finger variation and drift. Contribute to the Calibration and Sensing sections of the final paper.
Ali	Simulation and sim-to-real	Brought up the AmazingHand in physics simulation; loaded URDF/MJCF model and verified finger kinematics.	Implement multi-object force-awareness experiments (vary object stiffness and weight). Contribute to the Simulation section of the paper.
Sidath	VR teleoperation	Built and deployed the Quest 2 APK. Established ROS 2 communication on Ubuntu and demonstrated live controller position and orientation streaming as ROS 2 topics.	Bind the VR controller pose to a robot arm for partial VR-driven pick-and-place

Thomas	Vision-based finger pose estimation	Built a webcam-based hand-pose pipeline using MediaPipe Hands.	Harden the pipeline against lighting and partial occlusion, and explore ways to get thumb abduction.

Future Milestones (Weeks 7-10)

With the MVP complete, the remaining weeks focus on three threads: (1) extending the force-aware grasping behavior to a broader set of objects, (2) closing the loop on teleoperation, and (3) producing the final deliverables of a written paper, a demo video, and a public GitHub repository. Each milestone below has a defined owner, an expected completion week, and a concrete acceptance criterion.

Milestone	Owner	Target	Acceptance Criterion
Multi-object force-awareness experiments	Ali	End of W8	Force-aware grasp validated on at least two different objects
Sim-to-real groundwork: object-rotation training in sim	Ali	End of W8	A simple in-hand object-rotation policy trained in simulation; rollouts logged. Real-hardware transfer attempted, with results (success or characterized failure) documented.
VR controller ➜ robot arm end-effector control	Sidath	End of W8	Using Valve Index controllers for per-finger tracking.
Per-finger calibration coefficient table	Shree	End of W8	All four fingers individually calibrated with their own power-law coefficients.

Hand-pose pipeline hardening	Thomas	End of W8	Pipeline tested under conditions. Range of motion improved for stronger grasping.
Final paper draft (all sections)	Everyone	End of W9	Full draft of the project paper
Final demo video	Bryce + everyone	W10	Edited video showing the full system
Public GitHub repository	Bryce	W10	Top-level README with relevant files uploaded
Final presentation	Everyone	W10	Presentation covering MVP, reach-goal demos, and lessons learned. Each member presents their subsystem.

Revised Rest-of-Quarter Timeline

The Gantt below reflects what actually happened in Weeks 2-6 and what remains for Weeks 7-10.

Task	W2	W3	W4	W5	W6	W7	W8	W9	W10
System design & component selection	✓	✓							
PCB design & fabrication (Bryce)		✓	✓						
Hand assembly & sensor integration		✓	✓	✓					
Firmware: ESP32 USB serial (Bryce)			✓	✓					

FSR calibration procedure (Shree)			✓	✓	✓				
Per-finger calibration extension (Shree)						■			
Simulation bring-up (Ali)		✓	✓	✓					
Testing & force-aware grasping				✓	✓				
MVP completion & validation					✓				
Initial Hand-pose pipeline (Thomas)				✓	✓				
Hand-pose hardening (Thomas)						■	■		
VR pose streaming over ROS 2 (Sidath)				✓	✓				
VR ➜ robot arm end-effector (Sidath)						■	■	■	
Multi-object force awareness (Ali)						■			
Sim-to-real object rotation (Ali)							■	■	
Paper writing (Everyone)						■	■	■	■
Demo video (Bryce + everyone)									■
GitHub repo finalization (Bryce)								■	■
Final presentation (Everyone)									■

Summary

The Tactile Hands MVP is complete. We have a sub $200 four-fingered robotic hand with calibrated fingertip force sensing (1N), driven by a custom ESP32-based prototype board, performing force-aware grasp-and-release of a fragile object. Three reach goals are partially demonstrated ahead of schedule (hand-pose mirroring, VR pose streaming, hand simulation). Weeks 7-10 will integrate these into a partial or fully teleoperated, arm-mounted system, extend the force-aware behavior to a broader object set, attempt sim-to-real for in-hand rotation, and produce the final paper and demo video.